

Assignment # 3

Due : 11/3/2025

1. *ElGamal Encryption*

Encrypt the following messages using ElGamal encryption ($\mathbb{Z}_{409}^*$ and $g = 19$):

- (a) Private key $x = 301$, ephemeral key $k = 23$, message $m = 59$.
- (b) Private key $x = 301$, ephemeral key $k = 135$, message $m = 59$.

Perform both encryption and decryption. Show every intermediate step of the ElGamal Scheme.

2. *ElGamal Encryption Concepts*

Considering the examples from problem 1, we see that the ElGamal encryption scheme is non-deterministic: A given message m has many valid encryptions.

- (a) Why is the ElGamal encryption scheme non-deterministic?
 - (b) How many valid ciphertexts are there for each message m (general expression)? How many are there for the system in problem 1 (numerical answer)?
 - (c) Consider the case that for two messages $m_1 \neq m_2$ the same ephemeral key $k_1 = k_2$ has been chosen for the ElGamal encryption. This kind of behavior occurs if no or a bad cryptographic PRNG is being used. Show how the message m_2 can be recovered from a known message ciphertext pair m_1, c_1 if the same k is used.
3. Implement the square and multiply algorithm using a computer language of your choice (I recommend sage, python, java, or other math environments such as magma. The environment should have native BigInteger support, so do not use C/C++ etc.). The program should print all intermediate results. For sage and python, you can use the following template:

```
def my_pow(b,e,m):
    """ Computes b^e mod m using the square and multiply algorithm"""
    x =pow(b,e,m) # remove this line and place your code here instead
    return x
```

Compute the following exponentiations $a^b \bmod N$ using your program:

- (a) $a = 235973, b = 456872884723247, N = 583884$
- (b) $a = 984327455683, b = 1253489582, N = 994348472542$

Print the output of your program (including the intermediate steps) and turn it in together with your source code.